

On Krylov Subspace Approximations to the Matrix Exponential Operator

Reproducing Results from Hochbruck & Lubich (1997)

Purple Team

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January 5, 2026

Outline

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Motivation

- **Goal:** Compute $e^{\tau A}v$ efficiently for large sparse matrices A .
- **Context:** Stiff ODEs; exponential integrators vs. linear solves $(I - \gamma\tau A)^{-1}v$.
- **Key Insight:** Krylov for $e^{\tau A}v$ converges faster; depends on $\mathcal{F}(A)$.

Arnoldi Process and Approximation

Krylov Subspace

$$\mathcal{K}_m(A, v) = \text{span}\{v, Av, \dots, A^{m-1}v\}.$$

Relation

$$AV_m = V_m H_m + h_{m+1,m} v_{m+1} e_m^T, \quad H_m = V_m^* A V_m \text{ (Hessenberg)}.$$

Approximation

$$e^{tA} v \approx \beta V_m e^{tH_m} e_1, \quad \beta = \|v\|_2.$$

Error Bound (Lemma 1)

For convex $E \supset \mathcal{F}(A)$, error:

$$\|e^{\tau A} v - \beta V_m e^{\tau H_m} e_1\| \leq \frac{M}{2\pi} \int_{\Gamma} |e^{\tau \lambda} - q_{m-1}(\lambda)| \cdot |\phi(\lambda)|^{-m} |d\lambda|.$$

- ϕ : Conformal map; decay vs. $e^{\tau \lambda}$ growth.

Case 1 (Theorem 2)

Eigenvalues in $[-4\rho, 0]$.

Bounds

- $\sqrt{4\rho\tau} \leq m \leq 2\rho\tau$: $\epsilon_m \leq 10e^{-m^2/(5\rho\tau)}$.
- $m \geq 2\rho\tau$: $\epsilon_m \leq 10(\rho\tau)^{-1}e^{-\rho\tau}(e\rho\tau/m)^m$.

Superlinear at $m \approx \sqrt{\rho\tau}$.

Results

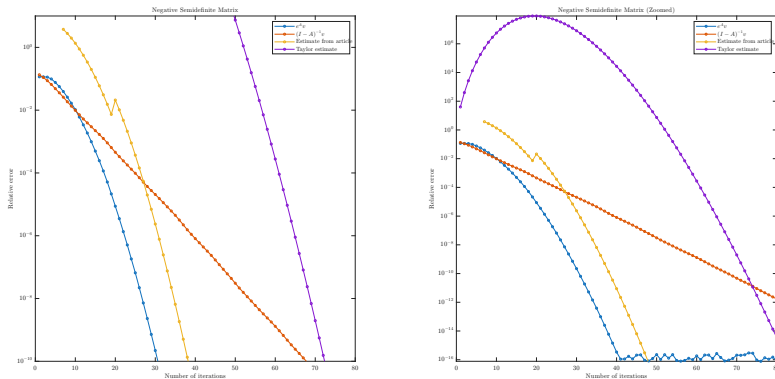


Figure: Error vs. CG for $(I - A)^{-1}v$; eigenvalues $[-40, 0]$, $N = 1001$.

Case 2 (Theorem 4)

Eigenvalues on imaginary axis, length 4ρ .

Bound

$$m \geq 2\rho\tau: \epsilon_m \leq 12e^{-(\rho\tau)^2/m}(e\rho\tau/m)^m.$$

Delay until $m \approx \rho\tau$.

Results

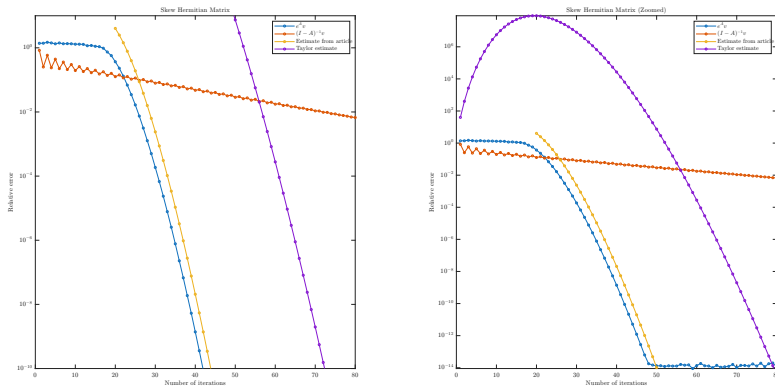


Figure: Error vs. BiCG for $(I - A)^{-1}v$; eigenvalues $[-20i, 20i]$, $N = 1001$.

Lanczos Approximation

- Biorthogonal V_m, W_m ; $H_m = D_m^{-1} W_m^* A V_m$ (tridiagonal for Hermitian).
- Matches Arnoldi for symmetric/skew cases.
- Bounds involve pseudospectra.

Results

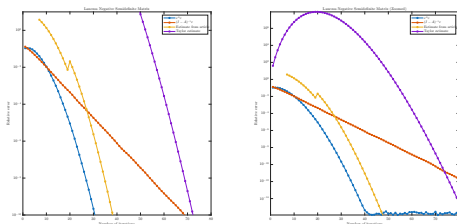


Figure: Symmetric negative semidefinite.

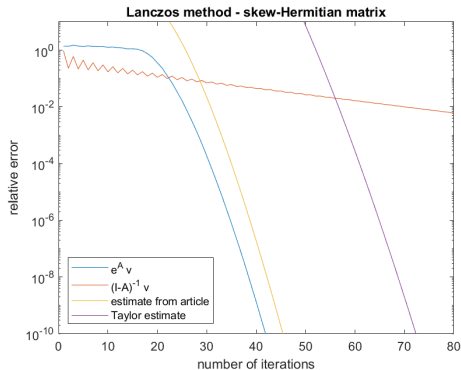


Figure: Skew-Hermitian.

Conclusion

- Reproduced bounds/plots for symmetric negative semidefinite and skew-Hermitian.
- Hierarchy: Faster for symmetric ($m \approx \sqrt{\rho\tau}$) than skew ($m \approx \rho\tau$).
- Exponential Krylov superior for stiff ODEs.

Reference

- M. Hochbruck and C. Lubich, *On Krylov Subspace Approximations to the Matrix Exponential Operator*, SIAM J. Numer. Anal., 34(5):1911–1925, 1997.